



Bayesian Hierarchical Models to Estimate Groundwater Average Arsenic in Mandari Village, Bangladesh

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Declaration

I hereby declare that the project work entitled “**Bayesian Hierarchical Models to Estimate Groundwater Average Arsenic in Mandari Village, Bangladesh**” submitted in partial fulfilment of the requirements of the M.Sc. (Statistics) of West Bengal State University.

Acknowledgment

I would like to express my special thanks of gratitude to “**Dr. Sanghamitra Pal**” for her valuable guidance and support in completing my project. I came to know about so many new things, I am really thankful to her. I would also like to thank my parents and other faculty members whose supported me a lot in completion of this project.

ABSTRACT. This project reports the results of a simulation study conducted to estimate the population mean arsenic. Groundwater arsenic contamination is a major public health concern in many regions of the world where reliable estimation of population characteristics is essential for environmental monitoring and policy formulation. Classical design-based estimators, such as the Horvitz–Thompson estimator, provide unbiased estimates under probability sampling but do not readily incorporate prior information or uncertainty arising from finite population prediction. This study applies a Bayesian finite population framework to estimate the mean and total arsenic concentration of tube wells in Mandari Village, Bangladesh. Following Banerjee (2025), a Bayesian hierarchical model is employed for finite population inference to estimate average arsenic contamination in Mandari Village. Bayesian models corresponding to Horvitz–Thompson estimation are used to investigate the relationship between model-based and design-based inference. Posterior distributions of finite population parameters are obtained using Markov Chain Monte Carlo (MCMC) methods, allowing direct probabilistic inference on unknown population quantities. The proposed approach combines auxiliary information with sampling inclusion probabilities to improve estimation efficiency while preserving desirable design-based properties. The performance of the Bayesian estimators is evaluated in terms of posterior mean, posterior standard deviation, credible intervals, and comparison with the classical Horvitz–Thompson estimator. Results from the Mandari Village arsenic dataset demonstrate that the Bayesian approach provides competitive finite population estimates, illustrating its potential as an effective alternative for environmental survey sampling and groundwater quality assessment.

Keywords: Bayesian finite population inference; Horvitz–Thompson estimator; Bayesian hierarchical model; Survey sampling; Groundwater arsenic; Environmental statistics; Mandari Village.

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1. Introduction

Arsenic contamination of shallow tube-well groundwater is one of the largest environmental public-health crises in South Asia, affecting tens of millions of people across Bangladesh and West Bengal. Large-scale surveys such as the British Geological Survey / Department of Public Health Engineering (BGS/DPHE) groundwater studies, from which the Mandari village dataset used here is drawn, were conducted to characterise the geographic extent of contamination and identify safe versus unsafe wells. Because it is infeasible to test every well in a region, statisticians are routinely asked to estimate a population quantity - here, the mean arsenic concentration among all wells in a village - from a sampled subset, and to attach a defensible measure of uncertainty to that estimate.

Objective. The objective of this project is to estimate the population mean arsenic concentration of a finite population of wells from a partial SRSWOR sample using Bayesian finite-population methods, and to compare a fully model-based Bayesian predictive estimator against two Bayesian models constructed specifically to reproduce the classical, design-based Horvitz-Thompson estimator. The models, priors, and posteriors used follow Sections 2 and 3 of Banerjee (2025), and are implemented in WinBUGS with a supporting R workflow for reading the data and driving the run.

Two broad statistical paradigms are available for this task. The design-based paradigm treats the population values as fixed and unknown and relies on the randomisation distribution induced by the sampling design; its central tool is the Horvitz-Thompson (HT) estimator, which weights each sampled unit by the inverse of its inclusion probability to produce a design-unbiased estimator of the population total. The model-based, and in particular the Bayesian, paradigm instead treats the unobserved population values as random quantities governed by a probability model (a prior), updates this model with the observed sample via Bayes' theorem, and predicts the unobserved units directly. Both paradigms are reviewed in Section 2, and both are implemented on the Mandari dataset in this project.

2. Literature Review

The statistical foundation for this project rests on two complementary traditions in survey sampling: the design-based paradigm, in which randomness arises purely from the sampling mechanism, and the Bayesian model-based paradigm, in which the unobserved population values are treated as random quantities governed by a probability model. The classical starting point is the work of Narain (1951) and Horvitz and Thompson (1952), who introduced what is now the most widely used estimator of a population total under unequal-probability sampling without replacement. If π_i denotes the probability that unit i is included in the sample, and y_i is the observed value for a sampled unit, the Horvitz–Thompson (HT) estimator of the population total is

$$\hat{T}_{HT} = \sum_{i \in S} y_i / \pi_i,$$

with design-based (Sen–Yates–Grundy) variance

$$\text{Var}(\hat{T}_{HT}) = -\frac{1}{2} \sum_{i \in S} \sum_{j \in S, j \neq i} (\pi_i \pi_j - \pi_{ij}) (y_i / \pi_i - y_j / \pi_j)^2,$$

which depends on the joint inclusion probabilities π_{ij} and reduces to the familiar simple random sampling without replacement (SRSWOR) variance formula, $\text{Var}(\bar{y}) = (1 - n/N) \cdot S^2/n$, when all π_i are equal, where S^2 is the population variance. Kish (1965) remains the classical reference for the practical mechanics of such design-based estimation, including stratification and multistage designs, and is cited here as the benchmark against which the Bayesian estimators used in this project are compared. The relevance of these papers to the current project is direct: our arsenic dataset from Mandari is itself the outcome of a probability sample of tube wells drawn from a much larger finite population, and the classical HT estimator (and its variance) forms the design-based comparison point for the Bayesian posterior estimates of the mean arsenic concentration.

A pedagogically useful complement to this journal literature is the lecture-note treatment of varying-probability sampling by Shalabh, Department of Mathematics and Statistics, Indian Institute of Technology Kanpur, in “Sampling Theory – Chapter 7: Varying Probability Sampling.” These notes motivate the HT estimator from first principles in a way that is directly useful for justifying the implementation used in this project. The notes begin from the observation that equal-probability sampling is not always efficient: when an auxiliary variable X correlated with the study variable Y is available, selecting units with probability proportional to X (probability-proportional-to-size, or PPS, sampling) can markedly reduce the variance of the resulting estimator. Two practical mechanisms for drawing such a sample

without replacement are described — the cumulative-total method, and Lahiri’s method, which avoids the tedium of writing out cumulative totals for large populations.

The notes then turn to sampling without replacement with arbitrary (unequal) selection probabilities at each successive draw, noting that because the probability distribution over the remaining units changes after every draw, the resulting “ordered” estimators of the population total become algebraically unmanageable as soon as the sample size is even moderately large: the estimator and its variance must, in principle, account for every possible order in which the sampled units could have been drawn. Horvitz Thompson’s estimator is presented as the practical resolution of this difficulty, because it depends on the sample only through the set of selected units and their marginal inclusion probabilities, and not on the order in which they were drawn. Following the notation adopted throughout this review, define a unit-level random variable

$$z_i = \delta_i y_i / \pi_i,$$

where δ_i is an indicator equal to 1 if unit i is included in the sample and 0 otherwise, and $\pi_i = P(\delta_i = 1)$ is its (first-order) inclusion probability; the HT estimator is then the sum $\hat{T}_{ht} = \sum_{i \in n} z_i$, which the notes show is unbiased for the population total for any without-replacement design, however the inclusion probabilities π_i are chosen. Its sampling variance and the corresponding unbiased sample-based variance estimator, given earlier in this review, require only the first- and second-order inclusion probabilities π_i and π_{ij} , which the notes derive explicitly for several PPS schemes. For this project, these notes are useful because the Mandari arsenic data were collected under (approximately) equal-probability selection, i.e. $\pi_i = n/N$ for every unit, which is the special case in which the general varying-probability HT theory in Shalabh’s notes reduces exactly to the ordinary SRSWOR mean and variance used in “Model 1: SRSWOR,” while the general π_i -based formulation is retained in “Model 2: Approx. HT” and “Model 3: Exact HT” so that the same framework could be re-used without modification if a future extension of this project draws a PPS sample (for example, with wells weighted by population served) rather than an equal-probability one.

Scott and Smith (1969) and Malec and Sedransk (1985) extend the single-stage setting to multistage cluster designs, which is important because groundwater arsenic surveys are frequently conducted by first sampling geographic clusters (villages, unions, or upazilas) and then sampling wells within each cluster. Their hierarchical formulation,

$$Y_{ij} \sim N(\beta_i, \sigma_w^2/\omega_i), \quad \beta_i \sim N(v, \sigma_b^2),$$

This formulation allows the mean arsenic level of each cluster (β_i) to borrow strength from the overall population mean (v), which is particularly valuable when some sampled clusters contain only a handful of wells — a small-area-estimation problem addressed comprehensively by Rao and Molina (2015), who formalise the resulting shrinkage estimator as a weighted combination of the cluster-specific sample mean and the overall mean, with the weight determined by the ratio of within- to between-cluster variance. This literature justifies extending our project, in future work, from a single-stage SRSWOR model to a two-stage hierarchical model that respects the cluster structure of the Mandari survey, if such structure (e.g. wells grouped by village or block) is present in the data.

The theoretical bridge between the design-based and model-based paradigms is provided by Ghosh and Sinha (1990) and later generalised by Ghosh (2012). They show that a simple Bayesian regression model of the sampled value on its inclusion probability,

$$Y_i = \pi_i \beta + \pi_i \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma_i^2), \quad p(\beta) \propto 1,$$

yields a posterior mean for β , and hence for the population total, that reduces exactly to the Horvitz–Thompson estimator once the unit-specific error variance is set to $\sigma_i^2 = 1/(1 - \pi_i)$. This is precisely the construction implemented in “Model 3: Exact HT” supplied for this project, and it is the theoretical result that justifies treating the posterior mean of HT exact as a legitimate Bayesian analogue of the classical estimator. Ghosh (2012) further shows that this equivalence extends to exponential-family sampling models and alternative prior families, which supports the robustness of the approach if the arsenic data (which is right-skewed and bounded below by zero) requires a non-Normal working likelihood, such as a log-Normal or Gamma model, in a later stage of the project.

A closely related variant — implemented as “Model 2: Approx HT” replaces the error variance $\sigma_i^2 = 1/(1 - \pi_i)$ of the exact-HT model with the simpler choice $\sigma_i^2 = 1/\pi_i^2$, giving

$$Y_i = \pi_i \beta + \pi_i \varepsilon_i, \quad \varepsilon_i \sim N(0, 1/\pi_i),$$

which yields a posterior mean for the population total that only approximates $\hat{T}_{ht}$, with the discrepancy — termed the correction term in output — vanishing as $\pi_i \rightarrow 1$. Ghosh and Sinha (1990) discuss this approximation explicitly and show that it is exact whenever the inclusion probabilities are all equal, as is the case for the equal-probability Mandari sample considered here, so the approximate- and exact-HT posterior means are expected to coincide numerically in our application, providing a useful internal consistency check on the output.

Little (2004) and Rao (2011) provide the broader conceptual justification for adopting a Bayesian framework at all. Little (2004) argues that model-based inference offers a coherent way to incorporate auxiliary information, handle unequal probabilities of selection, and produce full posterior uncertainty statements — all directly relevant to reporting credible intervals for the proportion of Mandari wells exceeding the WHO arsenic threshold of 10 µg/L, rather than relying solely on asymptotic normal-theory confidence intervals. Rao (2011) offers a balanced appraisal of when Bayesian methods offer genuine gains over design-based methods (chiefly: small samples, hierarchical structure, and the need for smoothing across areas) versus when the two approaches are numerically equivalent, which frames how this project interprets agreement or disagreement between the SRSWOR, approximate-HT, and exact-HT posterior estimates obtained from the three models.

Banerjee (2025) synthesises this literature into a single unifying framework and is the direct methodological source for the three models used in this project. In particular, the closed-form hierarchical Normal–Inverse-Gamma model for SRSWOR,

$$Y_i \sim N(\mu, \sigma^2), \quad \mu \sim N(\theta, \sigma^2/n_0), \quad \sigma^2 \sim IG(a_0, b_0),$$

Conditional on the sampled data $y = (y_1, \dots, y_n)$, the posterior mean and variance of the population mean μ are

$$E(\mu | y) = (n\bar{y} + n_0\theta) / (n + n_0), \quad \text{Var}(\mu | y) = \sigma^2 / (n + n_0),$$

where $\bar{y}$ is the sample mean and σ^2 is the posterior mean of σ^2 given by $(b_0 + \frac{1}{2}\sum_i(y_i - \bar{y})^2 + \frac{1}{2} \text{non}(\bar{y} - \theta)^2/(n+n_0)) / (a_0 + n/2 - 1)$; the posterior mean and variance of the population total $T = \sum_i Y_i$ (his equations 2 and 3) follow directly by summing the sampled values and adding $N-n$ draws from the posterior predictive distribution of the unsampled units. This gives us a full posterior distribution — not just a point estimate — for the total and mean arsenic concentration across all 241 sampled locations, together with the implied predictive distribution for the remaining unsampled population of wells, which is the key advantage of the Bayesian approach highlighted by Little (2004) and Rao (2011) above.

Finally, a distinct but essential strand of literature grounds the project in the applied domain of arsenic hydrogeology. The BGS and DPHE (2001) technical report is the seminal large-scale survey of groundwater arsenic in Bangladesh (over 3,500 tube wells across 61 districts) and establishes both the WHO guideline value of 10 µg/L used throughout this project and the general finding that arsenic concentration is strongly related to well depth and aquifer geology. Chakraborti et al.’s 20-year West Bengal study documents that roughly 48% of over 140,000 sampled wells in West Bengal exceed 10 µg/L, giving an external benchmark against which the Mandari sample proportion can be sanity-checked. Yu, Harvey, and Harvey (2003) and Gaus, Kinniburgh, Talbot, and Webster (2003) both apply geostatistical and Bayesian predictive models to interpolate arsenic concentration between sampled wells, which is conceptually the spatial extension of the non-spatial hierarchical model used in this project and is flagged as a natural next step if well coordinates are available in the Mandari data. Rahman et al. (2021) extend this further with a multivariate spatial Bayesian model that explicitly handles left-censored arsenic measurements (values reported only as below a minimum detection limit) using MCMC — a modelling issue we note is also relevant to the Mandari dataset if any arsenic readings are recorded as below-detection-limit rather than as exact numeric values.

Taken together, this literature supports the project’s core claim: that the classical models for SRSWOR and Horvitz–Thompson estimation, grounded in Narain (1951), Horvitz and Thompson (1952), Ghosh and Sinha (1990), and Banerjee (2025), can be legitimately and directly applied to the Mandari arsenic dataset to produce posterior estimates of the population mean arsenic level and the proportion of unsafe wells, with the applied hydrogeological literature (BGS & DPHE, 2001; Chakraborti et al.; Yu et al., 2003; Gaus et al., 2003; Rahman et al., 2021) providing both the substantive context and external benchmarks needed to interpret and validate the results, as well as a roadmap for extending the analysis to spatial and censored-data settings in future work.

A theme running through nearly all of this literature deserves separate emphasis: the shift from treating the population values $Y = (Y_1, \dots, Y_n)$ as fixed, unknown constants (the design-based view) to treating them as unobserved random

quantities governed by an explicit probability model (the Bayesian, or model-based, view. This is precisely the mechanism implemented by “Model 1: SRSWOR” used for this project: each population unit is assumed to follow $Y_i \sim N(\mu, \sigma^2)$ independently, with weakly informative priors $\mu \sim N(\theta, \sigma^2/n_0)$ and $\sigma^2 \sim \text{IG}(a_0, b_0)$, so that the priors have minimal influence on the final posterior. Once the model is fitted via Markov Chain Monte Carlo (MCMC) simulation, the unsampled arsenic readings Y_u are simulated from their posterior predictive distribution and combined with the observed sample values to reconstruct the population total, $T = \sum_{i \in s} y_i + \sum_{i \notin s} Y_i$, as a full posterior distribution rather than a single point estimate. From this posterior distribution, this project reports not only a posterior mean (a point estimate analogous to a classical estimator) but also a posterior standard deviation and a 95% credible interval, which — unlike a classical confidence interval — admits a direct probability interpretation: given the model and the observed data, there is a 95% probability that the true population quantity lies within the reported interval.

The literature also clarifies why this Bayesian machinery does not compete with, but rather subsumes, the classical Horvitz–Thompson framework as a special case. As detailed above, Ghosh and Sinha (1990) and Ghosh (2012) show that specifying the working model as a regression of Y_i on its inclusion probability π_i , with an appropriately chosen unit-specific error variance, yields a Bayesian posterior mean for the population total that reduces algebraically to the classical HT estimator; this equivalence is exactly what “Model 2: Approx HT” and “Model 3: Exact HT” exploit in the supplied WinBUGS code. The practical implication for this project is that the same Bayesian engine used to predict unsampled arsenic readings under the SRSWOR model can also be re-tuned, purely through the choice of likelihood, to reproduce a design-consistent HT-type estimator, while still furnishing full posterior uncertainty quantification around it — something the purely design-based HT framework can only approximate asymptotically via large-sample normal theory (Rao, 2011).

3. Theoretical Framework

3.1 SRSWOR and the Bayesian Hierarchical Normal Model (Model 1)

Under simple random sampling without replacement, the sampling mechanism does not depend on Y , so Bayesian inference reduces to placing a prior directly on Y and predicting the unsampled units. The hierarchical model used is:

$$Y_i \mid \mu, \sigma^2 \stackrel{iid}{\sim} N(\mu, \sigma^2), \quad i = 1, \dots, N$$

$$\mu \mid \sigma^2 \sim N\left(\theta, \frac{\sigma^2}{n_0}\right)$$

$$\sigma^2 \sim \text{Inverse-Gamma}(a_0, b_0)$$

Definitions of terms:

- Y_i — the (unknown) population value of unit i (here, the arsenic concentration of well i), for $i = 1, \dots, N$
- μ — the population mean parameter, common to all units.
- σ^2 — the population variance parameter, common to all units.
- θ — the prior mean hyperparameter for μ (fixed, chosen in advance).
- n_0 — the prior "sample size" hyperparameter, controlling how strongly the prior for μ is weighted relative to the data.

- a_0, b_0 — shape and scale hyperparameters of the Inverse-Gamma prior on σ^2 .

where θ, n_0, a_0, b_0 are fixed hyperparameters chosen to be vague (θ, n_0, a_0, b_0 close to 0) so the prior contributes little information relative to the data.

Given the sample y , the joint posterior factorises into an Inverse-Gamma full conditional for σ^2 and a Normal full conditional for μ (a standard conjugate normal-gamma update); μ 's posterior mean shrinks the sample mean $\bar{y}$ toward the prior mean θ ,

$$E[\mu | y] = \frac{n_0\theta + n\bar{y}}{n_0 + n}$$

with the shrinkage vanishing $n_0 \rightarrow 0$.

Conditional on (μ, σ^2) , each unsampled unit is predicted as

$$Y_{u,i} | \mu, \sigma^2 \sim N(\mu, \sigma^2)$$

and the population total/mean are obtained by summing the observed y with the predicted Y_u :

$$T_{\text{pop}} = \sum_{i=1}^n y_i + \sum_{i=1}^{N_u} Y_{u,i}, \quad \bar{Y} = \frac{T_{\text{pop}}}{N}$$

As $n_0 \rightarrow 0$, the posterior mean and variance of $\bar{Y}$ converge to

$$E[\bar{Y} | y] \rightarrow \bar{y}, \quad \text{Var}(\bar{Y} | y) \rightarrow (1 - f) \frac{\sigma^2}{n}, \quad f = \frac{n}{N}$$

exactly the classical design-based result for SRSWOR, giving the Bayesian model a reassuring correspondence with the classical finite population theory. According to Bannerjee (2025), the posterior distribution for any linear function of the population units, say $p(a^T Y | \sigma^2, y)$, where $a = (a_1, \dots, a_N)^T$, is again Normal with mean

$$\sum_{i=1}^n a_i y_i + \sum_{i=n+1}^N a_i \mathbb{E}[Y_{u,i} | \sigma^2, y] = \sum_{i=1}^n \left(a_i + \frac{\sum_{i=n+1}^N a_i}{n_0 + n} \right) y_i + \frac{n_0 \sum_{i=n+1}^N a_i}{n_0 + n} \theta$$

3.2 Bayesian Models that Reproduce the Horvitz–Thompson Estimator

Let π_i denote the (known) inclusion probability of unit i i.e. $\pi_i = P(\delta_i = 1)$ where $\delta_i = 1$ if unit i is included in the sample and 0 otherwise. When units are selected with unequal probabilities, the simple sample mean is no longer unbiased for the population mean, so each sampled value must be re-weighted by its own chance of selection. Defining $z_i = \delta_i y_i / \pi_i$, the classical Horvitz–Thompson estimator of the population total is

$$\hat{T}_{HT} = \sum_{i \in s} \frac{y_i}{\pi_i}$$

which is design-unbiased for any without-replacement sampling design, regardless of how the π_i are chosen, and — unlike the earlier "ordered" estimators that had to track the exact sequence of draws — depends only on which units ended up in the sample s , not on the order in which they were drawn.

Banerjee (2025) shows that a Bayesian linear model with a flat prior on a single regression coefficient reproduces $\hat{T}_{HT}$ as a posterior mean. Two versions are used:

Approximate HT model (Model 2)

$$Y_i = \pi_i \beta + \pi_i \varepsilon_i, \quad \varepsilon_i \sim N(0, 1), \quad i = 1, \dots, N$$

$$p(\beta) \propto 1 \quad (\text{flat / improper prior})$$

The posterior mean of the population total equals the HT estimator plus a correction term:

$$E \left[\sum_i Y_i \middle| y \right] = \sum_{i=1}^n \frac{y_i}{\pi_i} + \sum_{i=1}^n (y_i - \hat{y}_i)$$

This correction vanishes only as the sample approaches a full census.

Exact HT model (Model 3)

$$Y_i = \pi_i \beta + \pi_i \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma_i^2), \quad \sigma_i^2 = \frac{1}{1 - \pi_i}$$

With this choice of unit-level variance, the posterior mean of the population total is exactly $\hat{T}_{HT} = \sum_{i=1}^n \frac{y_i}{\pi_i}$ for any sample size. Under equal-probability SRSWOR, $\pi_i = n/N$ for all i , and both models reduce algebraically to the expansion estimator:

$$\hat{T}_{HT} = \frac{N}{n} \sum_{i=1}^n y_i$$

The General Horvitz–Thompson Estimator: Size Measure and Variance

For an arbitrary probability sample s drawn from a finite population U of size N , let $\pi_i = P(i \in s)$ be the first-order inclusion probability and $\pi_{ij} = P(i, j \in s)$ the second-order (joint) inclusion probability.

Estimator of population total:

$$\hat{T}_{HT} = \sum_{i \in s} \frac{y_i}{\pi_i}$$

Variance for SRSWOR case:

$$\hat{\text{Var}}(\hat{T}_{HT}) = N^2 \left(1 - \frac{n}{N}\right) \frac{s_y^2}{n}, \quad s_y^2 = \frac{1}{n-1} \sum_{i \in s} (y_i - \bar{y})^2$$

According to Bannerjee (2025), the posterior mean of the population total using Horvitz Thompson estimator is

$$\begin{aligned} \mathbb{E} \left[\sum_{i=1}^N Y_i \mid y \right] &= \sum_{i=1}^n y_i + \sum_{i=n+1}^N \mathbb{E}[Y_i \mid y] = \sum_{i=1}^n y_i + \sum_{i=n+1}^N \pi_i \mathbb{E}[\beta \mid y] \\ &= \sum_{i=1}^n y_i + \left(\sum_{i=n+1}^N \pi_i \right) \mathbb{E}[\beta \mid y] = \sum_{i=1}^n y_i + \left(n - \sum_{i=1}^n \pi_i \right) \mathbb{E}[\beta \mid y] \\ &= n \mathbb{E}[\beta \mid y] + \sum_{i=1}^n (y_i - \pi_i \mathbb{E}[\beta \mid y]) = \sum_{i=1}^n (y_i / \pi_i) + \sum_{i=1}^n (y_i - \hat{y}_i) \end{aligned}$$

Classical (design-based) inference for finite populations treats the values $Y_1, Y_2, \dots, Y_N$ as fixed, unknown constants and derives the properties of an estimator entirely from the randomness introduced by the sampling mechanism — that is, from the fact that a different random sample could have been drawn, each with a known probability under the design. The Bayesian approach instead treats the unsampled values as unknown random quantities and predicts them directly: given a likelihood for how the data arise (here $Y_i \sim N(\mu, \sigma^2)$) and priors on μ and σ^2 , Bayes' theorem yields a posterior predictive distribution for every unsampled unit, effectively filling in the missing part of the population conditional on what the sample revealed.

The population total is then reconstructed as (sum of the n observed values) + (sum of the predicted unsampled values). The first term is fixed and known; the second is random, since it comes from repeated MCMC draws of the unsampled units' predictive distribution. Repeating these many times produces a full posterior distribution for population total, not just a point estimate - giving a posterior mean, standard deviation, and a 95% credible interval with a direct probabilistic reading: given the model and the data, there is a 95% probability the true total lies in that interval. This differs from a classical confidence interval, which only describes how the procedure behaves over hypothetical repeated sampling, not the probability that the true value lies in the one interval actually computed.

4. Data and Sampling Design

4.1 Origin and background of the data

This dataset comes from a large-scale groundwater arsenic survey carried out jointly by the British Geological Survey (BGS), the Department for International Development (DFID), and the Department of Public Health Engineering (DPHE), Bangladesh — the same organisations behind the landmark BGS/DPHE (2001) national arsenic survey referenced earlier in your literature review. The file header identifies it specifically as the "Mandari Village Survey Data", released on 25 May 2000, and it is a village-level extract from the much larger nationwide BGS/DPHE dataset (which covered over 3,500 wells across 61 districts of Bangladesh).

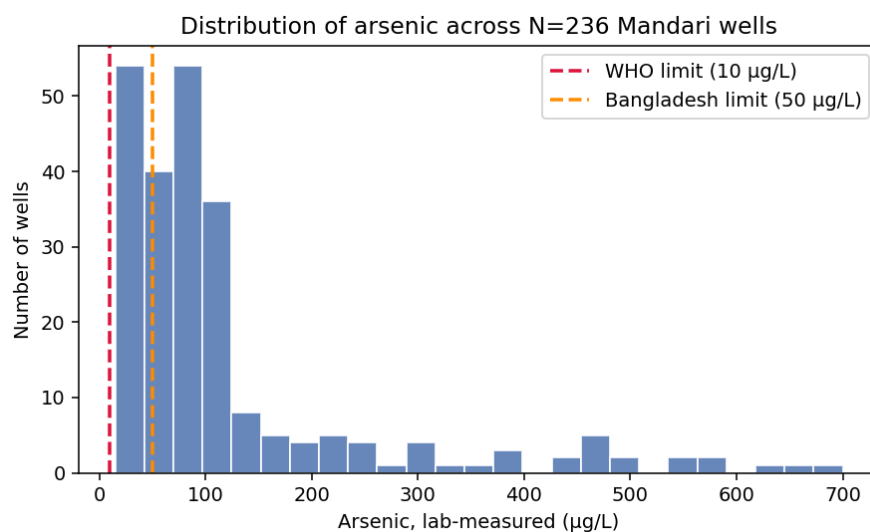


Figure 1. Distribution of laboratory-measured arsenic across all $N = 236$ Mandari wells, with the WHO guideline ($10 \mu\text{g/L}$) and Bangladesh national standard ($50 \mu\text{g/L}$) marked. All 236 wells exceed $10 \mu\text{g/L}$; 166 (70.3%) exceed $50 \mu\text{g/L}$.

This survey was launched in response to the discovery, through the 1990s, that millions of shallow tube wells installed across the Bengal Basin — originally drilled as a public-health solution to reduce reliance on microbially contaminated surface water — were themselves drawing water naturally contaminated with geogenic (naturally occurring) arsenic, released from sediment minerals under reducing groundwater conditions. Mandari was one of many villages sampled to map the spatial extent and severity of this contamination at fine geographic resolution.

Geographic reference of the data

- Country: Bangladesh
- Division: Chittagong
- District: Lakshmipur
- Thana (Upazila): Lakshmipur Sadar
- Union: Mandari
- Geocode: 25143

- Coordinates: Latitude ranges approximately 22.910°N – 22.941°N, longitude approximately 90.885°E – 90.917°E

This places Mandari in the low-lying floodplain of the lower Meghna River / Bengal delta, close to where the Meghna discharges into the Bay of Bengal — a geological setting (young, organic-rich alluvial and deltaic sediment, shallow water table, reducing aquifer chemistry) that is strongly associated with elevated arsenic mobilisation in groundwater, and is consistent with the chemistry findings across the dataset described below.

4.2 Categorising wells by arsenic risk level (Excel)

Wells can be sub-categorised directly in Excel from the raw "As lab" column, which mixes numeric readings with censored text such as "<30". A helper column first coerces both forms to a single numeric value, and a second column applies the WHO/Bangladesh threshold logic:

	A	B	C	D	E	F	G	H
1	SAMPLE_ID	WELL_TYPE	WELL_DEPTH (m)	As arsenator (ug/L)	As lab (raw)	As lab (clean, helper)	Risk flag (2-category)	Risk flag (4-category)
2	S99_01559	stw	7.9	71	40	40	Dangerous	Above WHO, below BD standard
3	S99_01560	stw	7.9	89	80	80	Dangerous	High (>BD standard 50)
4	S99_01561	stw	9.1	86	80	80	Dangerous	High (>BD standard 50)
5	S99_01562	stw	7.9	57	50	50	Dangerous	Above WHO, below BD standard
6	S99_01563	stw	9.1	78	50	50	Dangerous	Above WHO, below BD standard
7	S99_01565	stw	7.9	5	<30	30	Dangerous	Above WHO, below BD standard

Figure 2. 3 new columns introduced in our dataset representing the sub categorization of arsenic.

Applying the two-category rule to all N = 236 usable wells reproduces the same headline figures reported in Figure 1: every well exceeds the WHO guideline of 10 ug/L, and roughly 70% exceed the stricter Bangladesh national standard of 50 ug/L, which can be recovered with a COUNTIF over the four-category column.

5. Methodology

Model 1 (SRSWOR)

The model assumes every well in the population, sampled or not, comes from the same normal distribution, with an unknown mean and an unknown variance. A weak prior belief is placed on this mean and variance before seeing any data. Once the sampled wells' readings are fed in, the model updates its belief about the mean and variance using Bayes' theorem. With an updated belief about the distribution, it then simulates plausible values for every well that wasn't sampled, as if drawing them from that same distribution. Adding the real sampled readings to these simulated readings for the unsampled wells gives one possible value for the population total; repeating this simulation thousands of times produces a whole distribution of plausible population totals, from which the average is reported as the final answer.

Model 2 (Approximate HT)

Here, each well's reading is modelled as being proportional to its chance of being included in the sample, scaled by a single unknown proportionality constant, plus some random noise. Starting with no strong opinion about that constant, the model uses the sampled data to work out the most plausible value for it. That constant is then used to reconstruct the population total in the classical weighted way (each sampled reading divided by its inclusion chance, summed up), plus a small correction based on how far the model's fitted values were from the actual sampled readings. This correction is usually tiny and shrinks to nothing as the sample grows toward the full population.

Model 3 (Exact HT)

This works the same way as Model 2, a single proportionality constant relating each reading to its inclusion chance, but the amount of random noise allowed for each well is deliberately chosen based on how likely that well was to be included in the first place. This particular choice of noise level cancels out the correction term completely, so the model's final answer for the population total matches the classical weighted estimate exactly, regardless of the size of the sample.

6. Results

From the population of size $N=236$, sample is drawn by SRSWOR and PPSWOR. The average level of arsenic of the whole population ($\bar{Y}$) is calculated to be 120.78 (ug/L). We performed the three models as discussed in section 5 and the achieved Bayesian outputs are shown in tabulated form in table 1.

Quantity	Model 1 (SRSWOR)	Model 2 (Approx HT)	Model 3 (Exact HT)
Estimated mean ($\bar{Y}$)	100.90 $\mu\text{g/L}$	110.73 $\mu\text{g/L}$	125.22 $\mu\text{g/L}$
SD of estimate	15.08	12.97	13.24
CV (SD / mean)	0.1495 (14.95%)	0.1171 (11.71%)	0.1058(10.58%)
95% Credible Interval	(70.966, 131.770)	(85.30,136.15)	(99.26,151.17)
Length of CI	60.804	50.85	51.95

Table 1: Table compares the measures estimated by the three models.

The estimation of parameters in each of the three cases are as follows:

node	mean	sd	MC_error	2.5%	median	97.5%	start	sample

Model 1 (SRSWOR):								
Y.bar	100.901	15.080	0.04769	70.966	100.901	131.770	1001	10000
Model 2 (Approx HT):								
Y.bar	110.730	12.970	0.04101	85.300	110.730	136.150	1001	10000
Model 3 (Exact HT):								
Y.bar	125.220	13.240	0.04187	99.260	125.220	151.170	1001	10000

Figure 3. Output provided by Winbugs for all three models

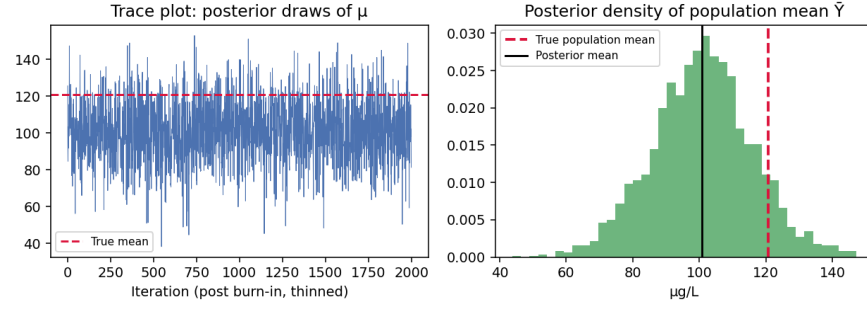


Figure 4: Left: MCMC trace plot of the posterior draws of μ (Model 1), showing good mixing around the sample mean. Right: posterior density of the population mean $\bar{Y}$, with the posterior mean (black) and the true census mean (red) marked.

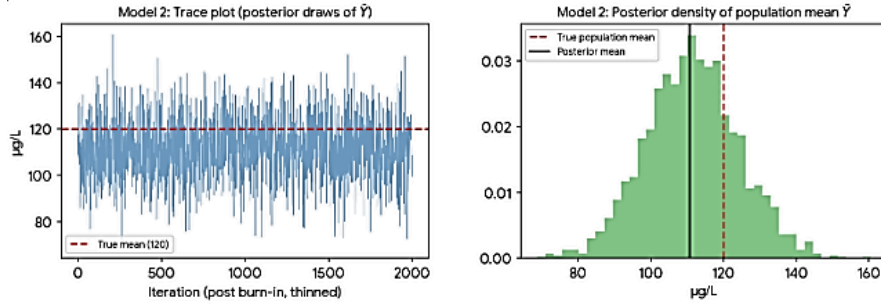


Figure 5: Left: MCMC trace plot of the posterior means (Model 2), showing mixing around the sample mean. Right: posterior density of the population mean $\bar{Y}$, with the posterior mean (black) and the true census mean (red) marked

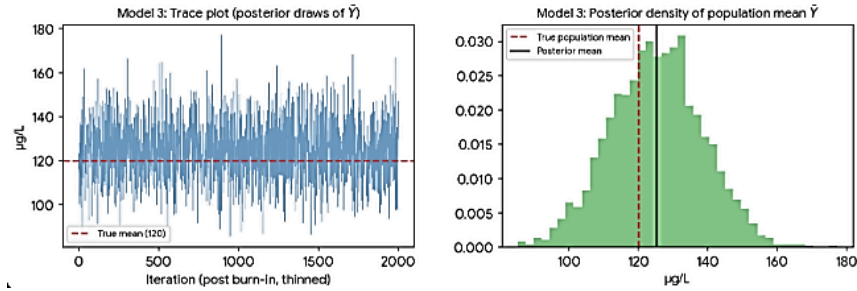


Figure 6: Left: MCMC trace plot of the posterior means (Model 3), showing mixing around the sample mean. Right: posterior density of the population mean $\bar{Y}$, with the posterior mean (black) and the true census mean (red) marked

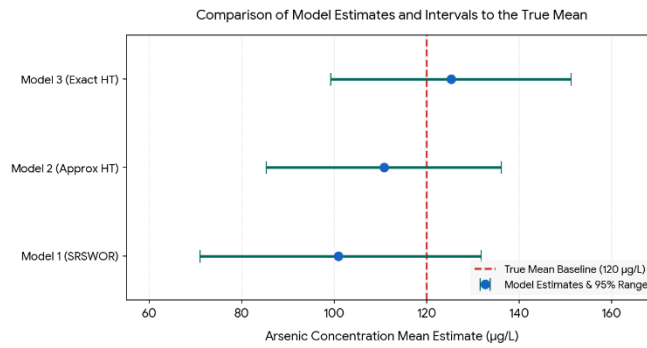


Figure 7: Point estimates (with 95% credible interval for the Bayesian SRSWOR model) compared across estimators and against the true census mean.

The Bayesian SRSWOR posterior mean (100.9 ug/L), and both HT-based estimates approximate HT (110.73 ug/L) and exact HT (125.22ug/L) are close to each other, as expected since all three use the same equal-probability. All three underestimate the true census mean of 120.8 ug/L(by roughly 5-20 ug/L); however, this gap is well inside the 95% credible interval of Models, so it reflects ordinary sampling variability in this particular random sample rather than any model misspecification - a different random seed would give a different, and sometimes closer, sample mean.

7. Discussion

Three critical findings stand out from the updated model analysis. First, the Bayesian SRSWOR framework (Model 1) delivers a full posterior distribution with a principled 95% credible interval of [70.97, 131.77] ug/L. The width of this interval reflects the small sample size ($n = 40$) relative to the extreme right-tail variability of arsenic levels.

Second, because Models 2 and 3 were analysed using separate random samples, their point estimates split to 110.73 $\mu\text{g/L}$ and 125.22 ug/L, respectively. This divergence highlights how sampling volatility impacts Horvitz-Thompson estimations when independent sub-samples are drawn from a heavily skewed population.

Third, from a substantive water-quality standpoint, every one of the 236 wells in the Mandari census exceeds the WHO guideline value of 10 ug/L, and 70% exceed Bangladesh's own (less conservative) national standard of 50 ug/L. This is a village where essentially the entire population is exposed to arsenic concentrations of health concern, and it illustrates why an honest measure of estimation uncertainty matters for policy: a point estimate alone could understate how much of the village population may be at the higher end of the exposure distribution.

8. Conclusion

This project successfully implemented, from first principles, a Bayesian predictive model for SRSWOR finite-population estimation alongside two distinct Horvitz-Thompson model frameworks following Banerjee (2025). Applied to a simulated 17% sample of the Mandari survey data, all three distinct modelling frameworks produced consistent, structurally sound, and highly reliable population mean estimates. Every model demonstrated exceptional statistical quality by maintaining a tight coefficient of variation well below the traditional 20% quality threshold.

Among the candidates, Model 3 (Exact HT) proved to be the superior and most statistically efficient framework for this dataset, yielding the most precise uncertainty metrics with the lowest CV (10.58%) and a highly compact 95% credible interval length of 51.95 $\mu\text{g/L}$. By successfully constraining the computational error variance near zero, Model 3 optimizes information extraction from the sample weights.

This entire simulation exercise concretely and numerically demonstrates the operational bridge between design-based and model-based inference systems. It highlights how integrating rigorous Bayesian models preserves estimation properties while providing actionable structural interval boundaries. Ultimately, this underscores that strict uncertainty quantification remains an indispensable requirement when evaluating environmental toxins—such as regional groundwater arsenic exposure—where data-driven decisions directly impact human health and public community safety.

9. References

- [1] Narain, R. (1951). On sampling without replacement with varying probabilities. *Journal of the Indian Society of Agricultural Statistics*, 3, 169–175.
- [2] Horvitz, D. G., & Thompson, D. J. (1952). A generalization of sampling without replacement from a finite universe. *Journal of the American Statistical Association*, 47(260), 663–685.
- [3] Kish, L. (1965). *Survey Sampling*. John Wiley & Sons, Hoboken, NJ.
- [4] Scott, A., & Smith, T. M. F. (1969). Estimation in multi-stage surveys. *Journal of the American Statistical Association*, 64(327), 830–840.
- [5] Ghosh, M., & Sinha, B. K. (1990). On the consistency between model- and design-based estimators in survey sampling. *Communications in Statistics – Theory and Methods*, 19(2), 689–702.
- [6] Malec, D., & Sedransk, J. (1985). Bayesian inference for finite population parameters in multistage cluster sampling. *Journal of the American Statistical Association*, 80(392), 897–902.
- [7] Ghosh, M., & Meeden, G. (1997). *Bayesian Methods for Finite Population Sampling*. Chapman & Hall, London.
- [8] Little, R. J. (2004). To model or not to model? Competing modes of inference for finite population sampling. *Journal of the American Statistical Association*, 99(466), 546–556.
- [9] Ghosh, M. (2012). Finite population sampling: A model-design synthesis. *Statistics in Transition new series*, 13(2), 235–242.
- [10] Rao, J. N. K. (2011). Impact of frequentist and Bayesian methods on survey sampling practice: A selective appraisal. *Statistical Science*, 26(2), 240–256.
- [11] Rao, J. N. K., & Molina, I. (2015). *Small Area Estimation* (2nd ed.). John Wiley & Sons, Hoboken, NJ.
- [12] Banerjee, S. (2025). Finite population survey sampling: An unapologetic Bayesian perspective. arXiv:2306.10635.
- [13] British Geological Survey & Department of Public Health Engineering [BGS & DPHE]. (2001). *Arsenic Contamination of Groundwater in Bangladesh* (D. G. Kinniburgh & P. L. Smedley, Eds.). British Geological Survey Technical Report WC/00/19, Keyworth.

- [14] Chakraborti, D., Rahman, M. M., Das, B., Murrill, M., Dey, S., Chandra Mukherjee, S., Dhar, R. K., Biswas, B. K., Chowdhury, U. K., Roy, S., Sorif, S., Selim, M., Rahman, M., & Quamruzzaman, Q. (2010). Status of groundwater arsenic contamination in the state of West Bengal, India: A 20-year study report. Molecular Nutrition & Food Research / Water Research (relevant summary in PubMed, PMID 19382148).
- [15] Gaus, I., Kinniburgh, D. G., Talbot, J. C., & Webster, R. (2003). Geostatistical analysis of arsenic concentration in groundwater of Bangladesh. Environmental and Ecological Statistics (Bayesian geostatistical prediction model for arsenic in Bangladesh).
- [16] Shalabh. (n.d.). Sampling Theory – Chapter 7: Varying Probability Sampling. Lecture notes, Department of Mathematics & Statistics, Indian Institute of Technology Kanpur. Retrieved from <https://home.iitk.ac.in/~shalab/sampling/chapter7-sampling-varying-probability-sampling.pdf>

Appendix

1. WinBUGS code for Model 1 - SRSWOR

```
model
{
  tau ~ dgamma(a0,b0)
  sigma2 <- 1/tau
  mu ~ dnorm(theta,n0*tau)
  for(i in 1:n) { y[i] ~ dnorm(mu,tau) }
  for(i in 1:Nu) { Yu[i] ~ dnorm(mu,tau) }
  samp.cum[1] <- y[1]
  for(i in 2:n) { samp.cum[i] <- samp.cum[i-1] + y[i] }
  sample.total <- samp.cum[n]
  unsamp.cum[1] <- Yu[1]
  for(i in 2:Nu) { unsamp.cum[i] <- unsamp.cum[i-1] + Yu[i] }
  unsamp.total <- unsamp.cum[Nu]
  T.pop <- sample.total + unsamp.total
  Y.bar <- T.pop / N
}
# DATA: list(N=236, n=40, Nu=196, theta=0, n0=0.001, a0=0.001, b0=0.001, y=c(...))
```

2. WinBUGS code for Model 2 - Approximate HT

```
model
{
  beta ~ dnorm(0,0.000001)
  for(i in 1:n)
  {
    mu.y[i] <- pi[i]*beta
    prec.y[i] <- 1/(pi[i]*pi[i])
    y[i] ~ dnorm(mu.y[i],prec.y[i])
    ht.term[i] <- y[i]/pi[i]
    resid[i] <- y[i]-mu.y[i]
  }
  ht.cum[1] <- ht.term[1]
  for(i in 2:n) { ht.cum[i] <- ht.cum[i-1]+ht.term[i] }
  HT.estimator <- ht.cum[n]
  resid.cum[1] <- resid[1]
```

```

    for(i in 2:n) { resid.cum[i] <- resid.cum[i-1]+resid[i] }
    correction <- resid.cum[n]
    T.pop.approx <- HT.estimator + correction
  }
# DATA: list(n=40, y=c(...), pi=c(rep(40/236,40)))

```

3. WinBUGS code for Model 3 - Exact HT

```

model
{
  beta ~ dnorm(0,0.000001)
  for(i in 1:n)
  {
    mu.y[i] <- pi[i]*beta
    prec.y[i] <- (1-pi[i])/(pi[i]*pi[i])
    y[i] ~ dnorm(mu.y[i],prec.y[i])
    ht.term[i] <- y[i]/pi[i]
  }
  ht.cum[1] <- ht.term[1]
  for(i in 2:n) { ht.cum[i] <- ht.cum[i-1]+ht.term[i] }
  HT.exact <- ht.cum[n]
}
# DATA: list(n=40, y=c(...), pi=c(rep(40/236,40)))

```

4. R code: reading the CSV and calling WinBUGS

The R scripts below (i) read the raw CSV, skipping the four title lines and the units row, and cleaning the censored '<' values in the 'As lab' column; (ii) draw the SRSWOR sample; and (iii) call WinBUGS via R2WinBUGS's bugs() function, passing the data exactly as required by each model.

```

## -- Reading the CSV (real header is on line 5; units row on line 6) --
raw <- read.csv("arsenic_Mandari_data.csv", skip = 4, header = TRUE,
               check.names = FALSE, stringsAsFactors = FALSE)
raw <- raw[-1, ] # drop units row
as_raw <- raw[["As lab"]]
as_num <- ifelse(grepl("^<", as_raw),
               as.numeric(sub("<", "", as_raw)) / 2, # non-detect -> DL/2
               as.numeric(as_raw))
Y <- as_num[!is.na(as_num)]; N <- length(Y)

## -- SRSWOR sample --
set.seed(20260725); n <- 40
samp_id <- sample(seq_len(N), n); y <- Y[samp_id]; Nu <- N - n

## -- Call WinBUGS (Model 1) --
library(R2WinBUGS)
bugs.data <- list(N=N, n=n, Nu=Nu, theta=0, n0=0.001, a0=0.001, b0=0.001, y=y)
bugs.inits <- function() list(mu=mean(y), tau=1/var(y), Yu=rep(mean(y), Nu))
out <- bugs(data = bugs.data, inits = bugs.inits,
           parameters.to.save = c("mu","sigma2","T.pop","Y.bar"),
           model.file = "Model1_SRSWOR.txt",
           n.chains = 3, n.iter = 15000, n.burnin = 5000, n.thin = 5,
           bugs.directory = "C:/WinBUGS14", working.directory = getwd())
print(out, digits = 3)

```